Aryan Alpesh Bhavsar

Computer Science & Engineering
Indian Institute of Technology Indore

2401201001
Ph.D. Candidate
Male

EDUCATION

Examination	University	Institute	Year	CPI/%
Doctor of Philosophy	IIT Indore	IIT Indore	2024-Present	8.88
Graduation	GTU	GIT Gandhinagar	2020-2024	8.09
Intermediate/+2	ISC	SGVP School	2018-2020	85.80
Matriculation	ICSE	SGVP School	2017-2018	80.33

ABOUT ME

I am a PhD student in Computer Science and Engineering at IIT Indore, working with Dr. Sidharth Sharma (IIT Jodhpur) and Dr. Ayan Mondal (IIT Indore). My research lies at the intersection of AI/ML systems and data center networking, focusing on building cyberinfrastructure and network-aware frameworks for large-scale distributed DNN training. I work on telemetry-driven optimization in data center networks, leveraging real-time network metrics such as latency, loss, and link utilization to adapt communication patterns, routing, multicast, and collective scheduling for distributed training workloads. My broader goal is to advance networks-for-AI, enabling scalable, efficient, and sustainable AI training and inference systems in large-scale data center and cloud environments.

RESEARCH EXPERIENCE & INTERNSHIPS

- **Research Scholar** | ***Indian Institute of Technology Indore*** *(09/2024 - Present)*
  - Reviewed literature on data center congestion patterns (microbursts, incast), multicast techniques, and low-latency transport concepts (including RDMA/RoCE), along with studies on distributed and geo-distributed ML training.
  - Built multiple Containerlab-based example topologies to develop a repeatable workflow for deploying and validating network configurations.
  - Implemented a symmetric RoCE-enabled IRB L3EVPN (EVPN-VXLAN) fabric in Containerlab using SR Linux switches.
  - Planning to extend the testbed with a centralised SDN controller and live telemetry collector.
- **Software Engineer Intern** | ***Jenex Technovation Private Limited*** *(01/2024 - 06/2024)*
  - Developed key modules of a cross-platform Flutter app, including authentication, role-based access, session persistence, form validation, and backend integration.
  - Implemented secure backend features in Node.js/MongoDB: image upload pipeline (Multer) with storage handling and API integration.
  - Built token-based authentication middleware supporting role-based authorization and protected API routes.
  - Implemented account workflows: registration email notifications, OTP-based password reset.
  - Integrated device connectivity using Android core APIs (Bluetooth) to interface with an IoT hardware component.
  - Debugged and improved existing web-app APIs.

PROJECTS

- **RoCE-enabled L3EVPN over EVPN-VXLAN in a Spine–Leaf Fabric (Containerlab + Nokia SR Linux)**
  - Implemented EVPN-VXLAN L3 connectivity with multiple VNIs- validated host reachability across leafs and segmentation across VNIs.
  - Configured IRB (Integrated Routing & Bridging) to enable L2/L3 gateway functionality and validated inter-subnet reachability.
  - Verified control-plane advertisements and end-to-end forwarding.
  - Structured BGP neighbour groups and VTEP settings.
  - Enabled ECMP in the fabric and validated fast failover: on link failure, BFD-assisted detection reduced convergence time.
  - Implemented and validated soft-RoCE (RDMA over Converged Ethernet) with in virtual machines hosted inside containerized environments (Containerlab).

- **Deepfake Audio Detection with Statistical Classifiers** - Pattern Recognition Course Project
  - Dataset used for training & cross validation- Fake or Real (FoR).
  - Compared multiple statistical classifiers and a combination of them(ensemble) - Logistic Regression model outperformed with a stable accuracy between both classes.
  - All models were also compared with same models trained on reduced dataset - Achieved higher accuracy in terms of Real Audio detection but with higher degradation in terms of Fake Audio detection.
  - SVM model shows human-biasness, achieving highest accuracy for Real Audio detection and lowest for Fake Audio detection.
- **ChatINC**
  - A web-based real-time chat application, made with MERN stack and tailwind CSS.
  - This project uses socket io, to provide the realtime messaging functionality to users.
  - Users are authenticated with a token-based middleware and cookies are used to save login information.
- **GAMEHUB**
  - A game displaying, responsive website with infinite scrolling, made with React + Typescript & Chakra UI.
  - Game cards can be sorted and searched based on platforms, genres and more.
  - React query is used to fetch data from rawg API, and caching is implemented to prevent recurring requests.
  - React router is used to navigate the web app and handle errors.

TECHNICAL SKILLS

- **Tools**: Network Emulation with ContainerLab, Docker, SR Linux (NOKIA) Configuration
- **Programming & Scripting Languages**: Java, Javascript, Python, Dart, Typescript
- **Development Technology**: MERN, MEAN, Flutter
- **Language Known:** English, French & Hindi

POR & ACHIEVEMENTS

- **Teaching Assistant** for the course, **CS334/434/634: Wireless Networks and Applications**, conducting tutorials for the course. *(01/2026 - Present)*
- **Teaching Assistant** for the course, **CS653: Programming Lab**, supervised the labs for the course. *(06/2025 - 11/2025)*
- **Teaching Assistant** for the course, **CS306: Computer Network**, supervised the labs for the course. *(01/2025 - 05/2025)*
- **AIR 2308** in **GATE CSE 2024** exam- *Marks: 51.09, Score: 606* *(02/2024)*
- **NPTEL certification** in **Python for Data Science**- 76% *(07/2022 - 08/2022)*
- **NPTEL certification** in **Joy of Computing With Python**- 83% *(07/2021 - 10/2021)*

CONTACT INFORMATION

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